Small Ideas, Smart Solutions:

Real-Time Alerts for Oracle APEX Interactive Reports using WebSockets



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The scenario

Business context

The client had a service request management system, with a report displaying requests categorized by priority.

Business requirement

Automatic report refresh was rejected by the business team, but users still needed to be notified when new requests became available.

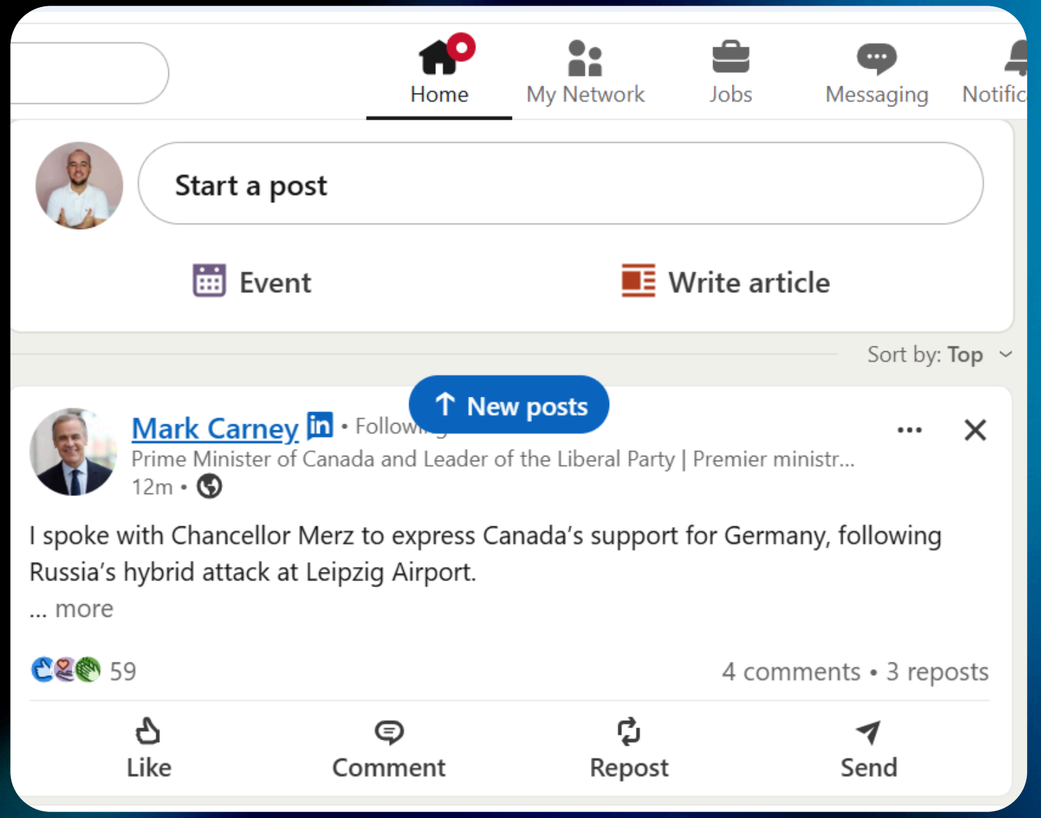
The challenge

Create a real-time alert that would draw users' attention to new data without refreshing the page.

The idea

Implement a notification button at the top of the interactive report, inspired by LinkedIn's notification indicator for new posts.

↑ New posts



The javascript solution. It works, but...

Too much workaround..."gambiarra"

- Use a JavaScript timer (setInterval) to trigger an AJAX callback every three minutes.
- Query the database to check whether new service requests had been created within the last five minutes.
- Use an APEX_COLLECTION to track previously notified requests and prevent displaying the same alert multiple times.

Polling method

Client sends HTTP requests to the server. Server responds. After a predermined time interval, the client sends a new request.

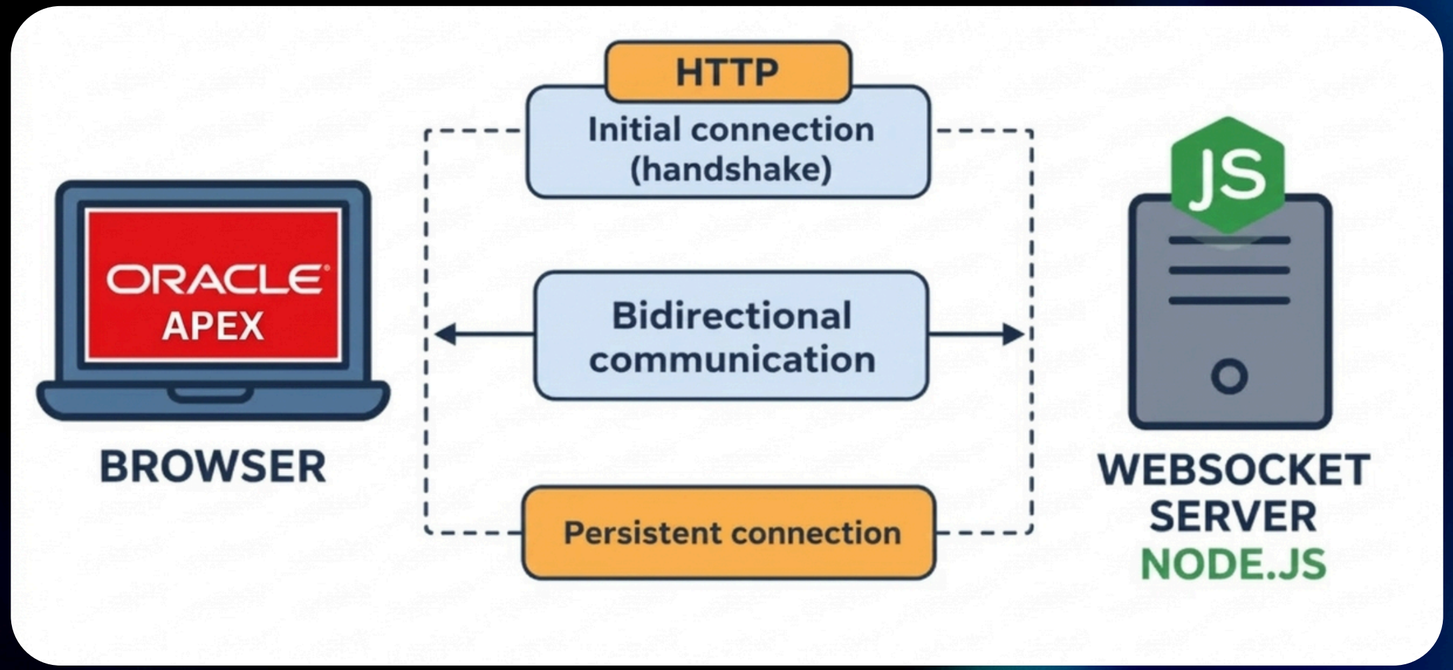
The WebSockets solution

What is WebSockets

WebSockets enable real-time, bidirectional communication between a client and a server over a single, persistent connection, allowing data to be exchanged without repeatedly sending requests.

Where is it most used

WebSockets are commonly used in chat applications, live dashboards, online gaming, financial market updates, collaborative tools, and real-time notification systems.



HTTP upgrade

The browser sends an HTTP request to establish the connection.

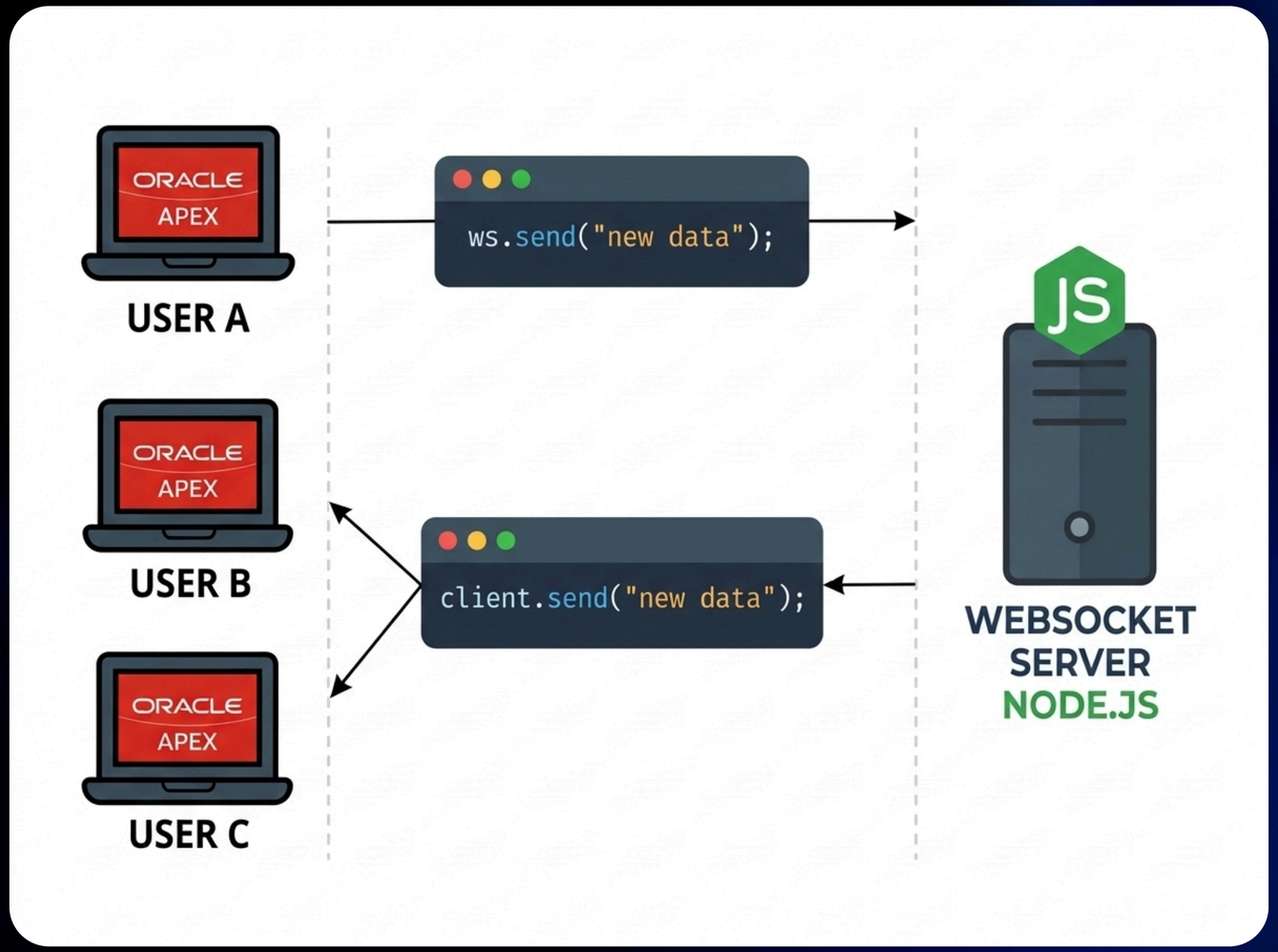
Handshake response

The server accepts the upgrade and the connection becomes a WebSocket.

Bidirectional communication

Both sides can send messages independently. The connection remains open until it is closed.

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It gets better with Socket.IO :)

Socket.IO is a library that makes real-time communication easier to implement.

Built on top of real-time communication

WebSocket provides the communication channel.

Socket.IO provides additional tools and abstractions that make using that channel easier.

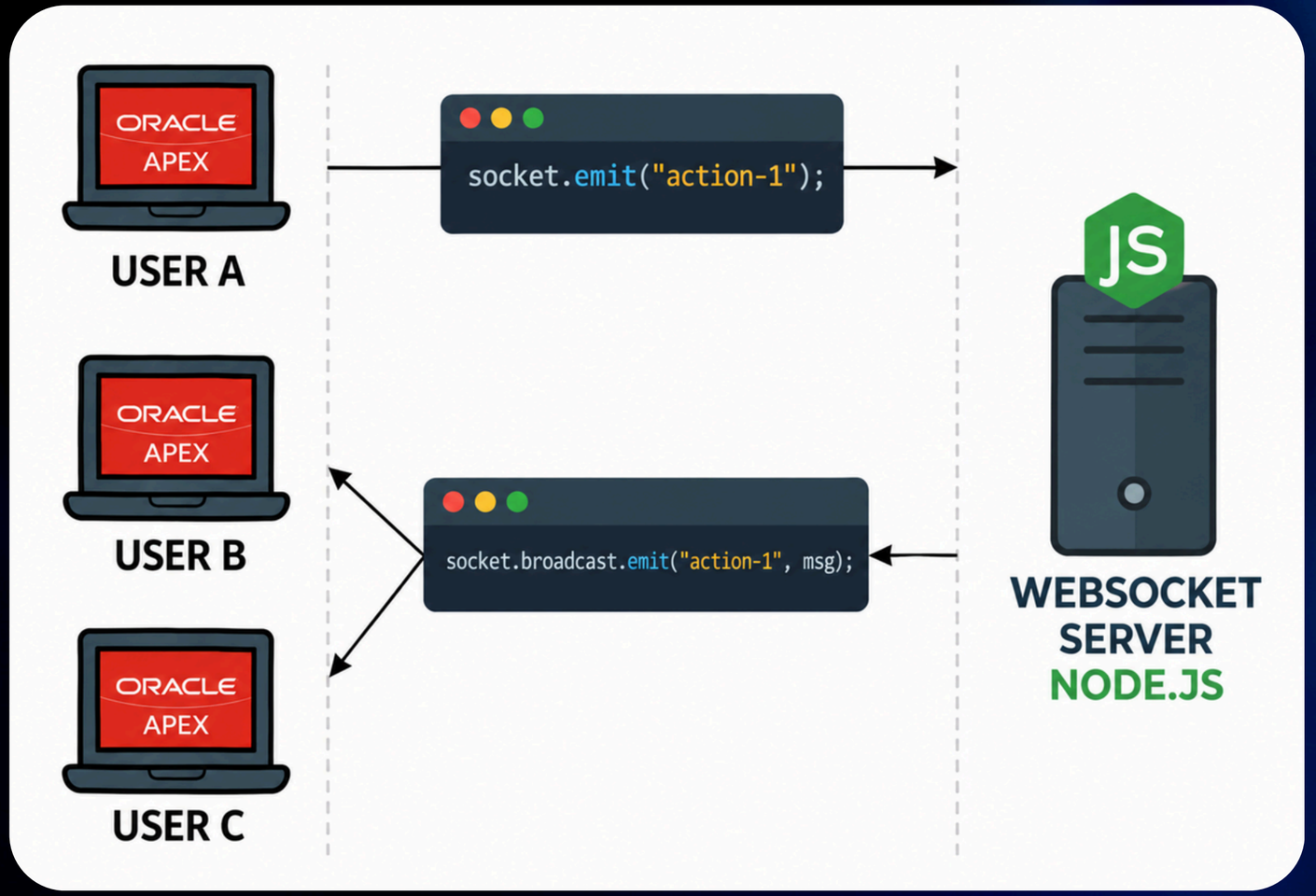
Simplifies communication

With WebSockets you send messages directly, with Socket.IO you can define named events.

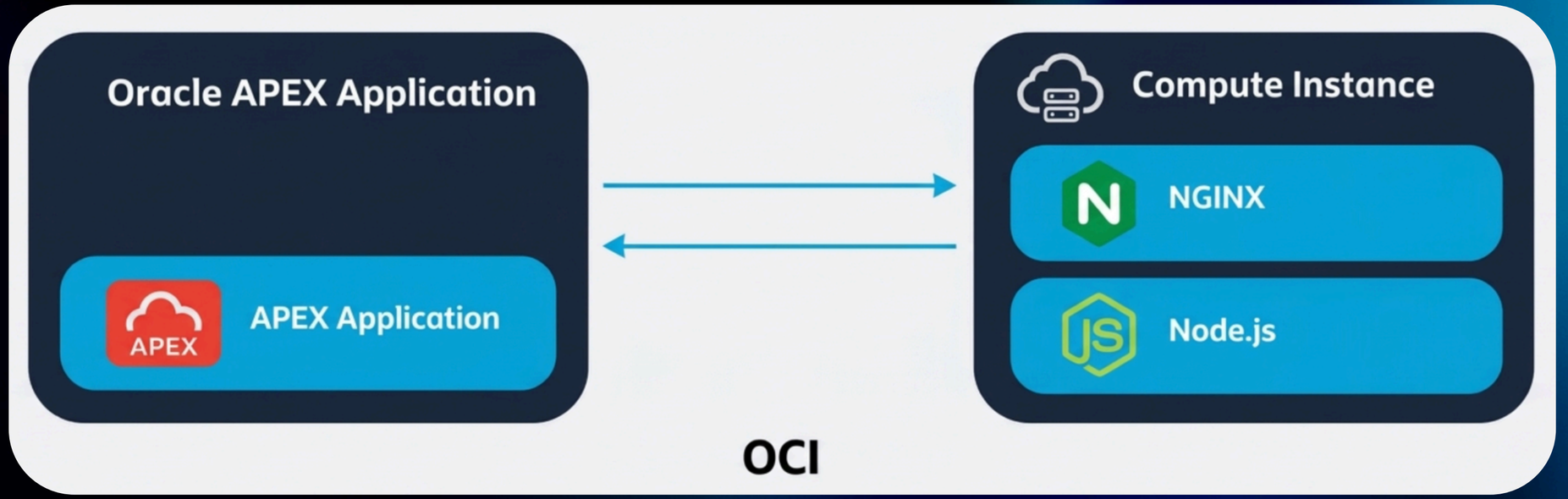
Adds useful features

Unlike raw WebSockets, Socket.IO provides features like automatic reconnection, event-based messaging, rooms, and broadcasting.

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Setup


Oracle APEX Application
APEX
APEX Application
Compute Instance
NGINX
Node.js
OCI

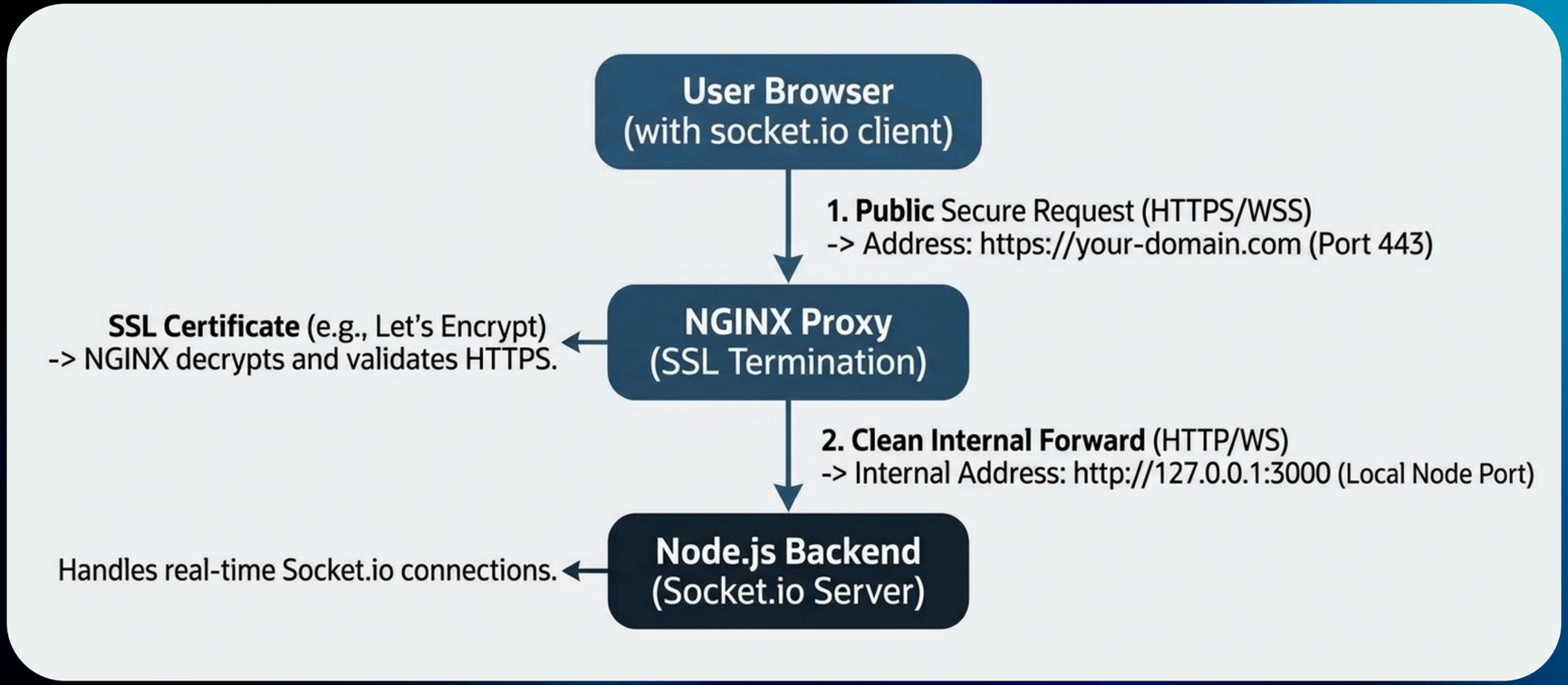
Demo

NGINX

Reverse Proxy & SSL

Node.js cannot manage SSL certificates efficiently and must run privately on port 3000.

NGINX handles this encryption at the edge to prevent Mixed Content Errors in your secure application.



It is worth it?

Thanks!

Any questions?

Try it yourself.

Your feedback.



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